

Random Variables

- Allow us to pass experimental outcomes to a numerical function
- Consider a random experiment with a sample space C . A function X , which assigns each element $c \in C$ exactly one $X(c) = x$, is called a random variable. The space or range of X is the set of real numbers $D = \{x: x = X(c), c \in C\}$.
- They are random because they take on random values based on the outcome of an experiment

Example

A fair die is thrown twice. The sample points are $(1,1), (1,2), \dots, (6,6)$. There are 36 sample points. Let us assign the same probability $1/36$ for each of these points.

Suppose we are interested in the sum of the numbers of each outcome. Then we can define a random variable $X = i+j$ associated with the outcome (i,j) .

List the possible values of X and find the probability of each value.

Value of X	2	3	4	5	6	7	8	9	10	11	12
Probability	$1/36$	$2/36$	$3/36$	$4/36$	$5/36$	$6/36$	$5/36$	$4/36$	$3/36$	$2/36$	$1/36$

Types of Random Variables

- Discrete $\Rightarrow$ Countable sets
- Continuous $\Rightarrow$ Uncountable sets